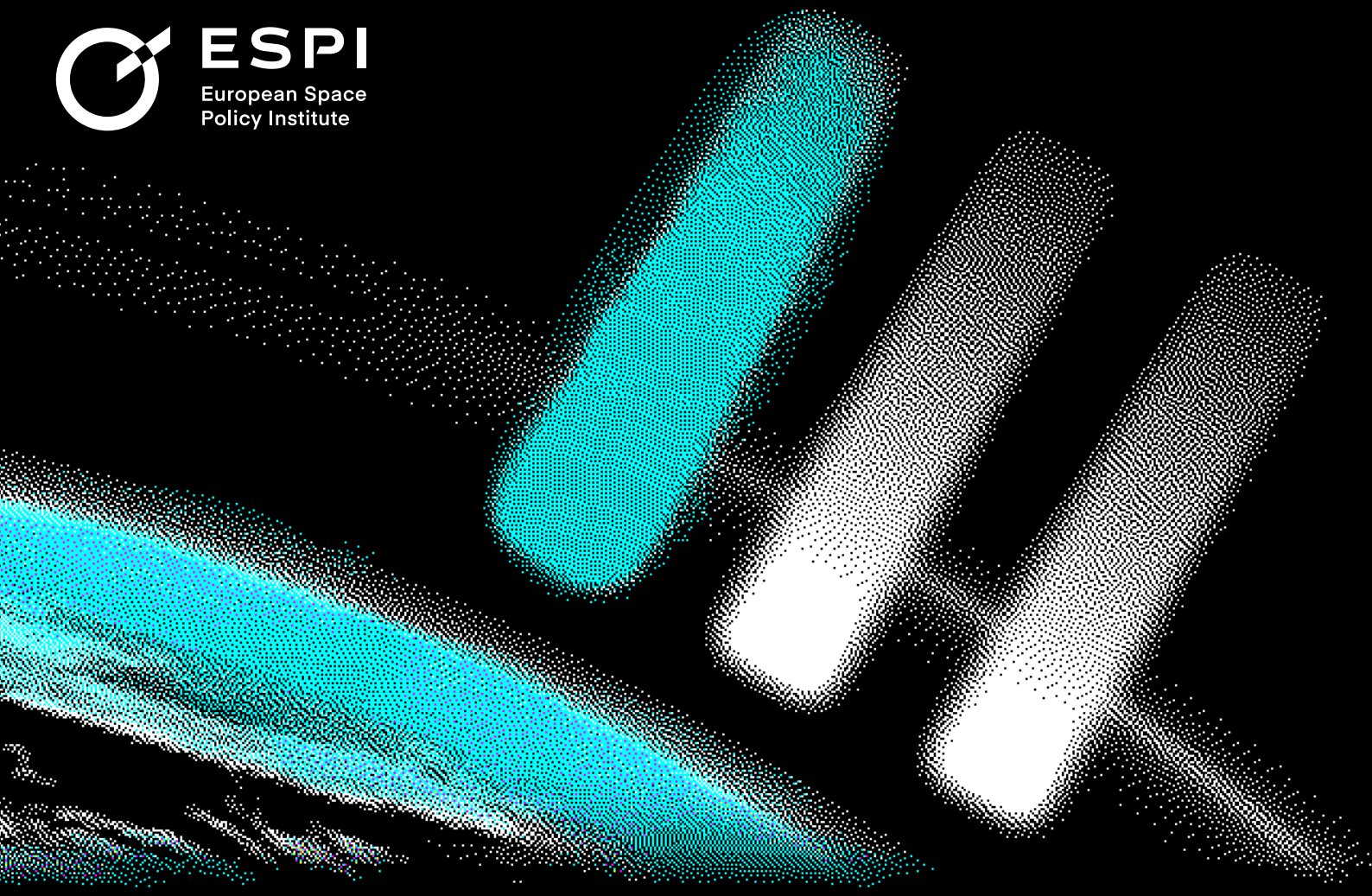
ESPI
European Space
Policy Institute



A POLICY VISION FOR EARTH OBSERVATION

FOOD, WATER, ENERGY, DEMOCRACY, PEACE EMPOWERED THROUGH SPACE





EVERY HEADLINE IS CLEARER FROM SPACE

A NEW ERA FOR EUROPEAN POLICY...

WHY A POLICY VISION FOR EARTH OBSERVATION?

Europe is standing at the cusp of a new era. An era where our weight in the world is measured by our ability to set our priorities, seize opportunities, anticipate crises, and act at scale with purpose. An era where we either shape the future or observe on the sidelines. An era where emerging and advanced technologies will determine our long-term prosperity and ability to ensure peace.

A new era requires a new approach, empowered by space. At ESPI, we have embarked on a new journey to propose clear policy guidance & pull to the space sector, complementing the technological and scientific foundations & push that have propelled our space programmes to date. For this reason, we convened, together with the European Space Agency, a high-level workshop on the Policy Value of Earth Observation.

At a time when security, climate resilience, and resource management issues rise to the top of policy agendas across Europe, Earth Observation has a much bigger role to play than ever before. Only with full integration of space in its sectoral policymaking will Europe deliver in domains such as energy, security, environment, agriculture, digitalisation and diplomacy. This Vision marks only the beginning of a decade-long effort to put space at the full service of Europe's values and goals.



H. Ludwig Moeller
Director of the European Space Policy Institute (ESPI)

...THE NEXT CHAPTER FOR SPACE AND EARTH OBSERVATION

This Vision expresses ESPI's views stemming from its own research and crucially informed by the outcome of a high-level Workshop on the Policy Value of Earth Observation, convened by ESPI and the European Space Agency (ESA) at the Royal Society, London, in April 2025. Selected quotes by the participants are featured throughout the document. This Vision also builds on further consultations, leveraging ESA's long-standing expertise in Earth Observation.

ABOUT

This Vision sets a direction, initiating a dialogue to maximise the policy impact of Earth Observation: bringing new energy to space programmes through stimulating continuous policy demand and guidance. It is addressed to national and European decision-makers in the domains of energy, security, environment, agriculture, digitalisation and beyond.

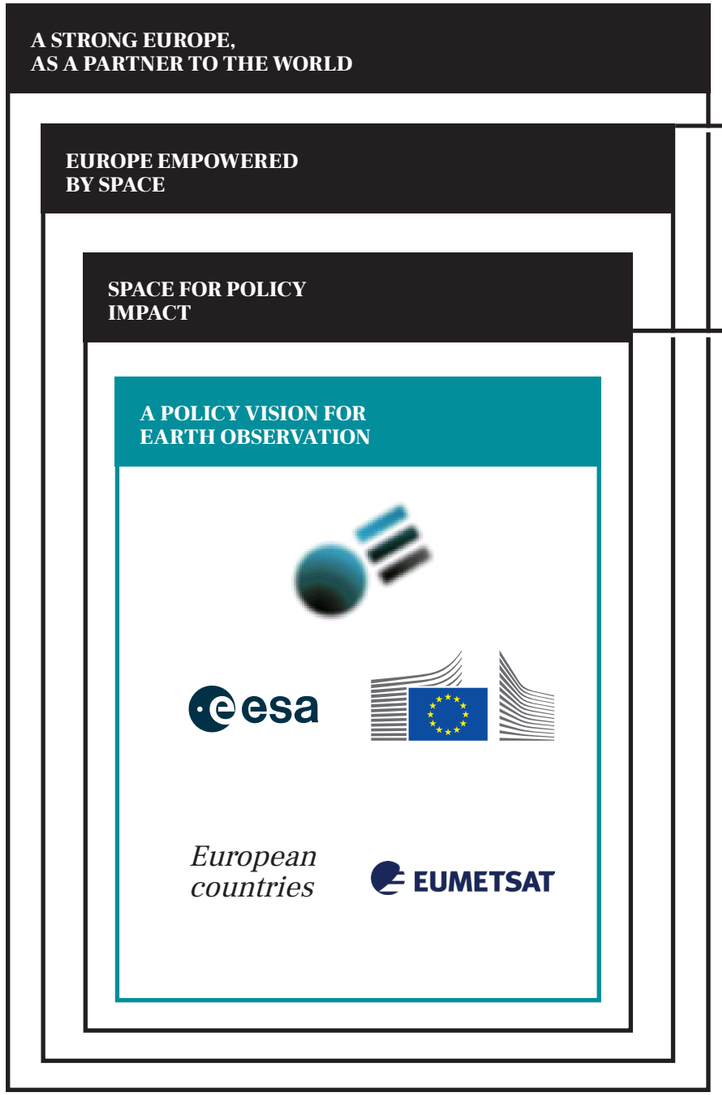
While the impact of space extends further, this Vision focuses only on the highest-priority domains and policy objectives where a stronger integration of Earth Observation would deliver significant added value to Europe. Collaboration between policy and space communities will unlock new opportunities, to date unexplored, that can bring value to European citizens, public administrations, and businesses through space-derived information and services.

While this document is not a work plan, the Vision is underpinned with a tangible call for action. It urges the policy community to further integrate Earth Observation, and space at large, into their sectoral policies and calls on the space community to create entry points to receive policy guidance for their programmatic action. 'It takes a village', in the spirit of the African proverb.

Doing so is no easy feat and represents a task of a decade. Success depends on timely action, including leveraging upcoming milestones that will shape the future of Earth Observation in Europe. Notably, the ESA Council Meetings at Ministerial Level and the parallel preparation of the new EU Multi-Annual Financial Framework for the 2028-2034 time frame, as well as the breadth of national initiatives.

THE VISION IN THE BIGGER PICTURE

This Vision fits within a broader context and contributes to a global positioning of Europe as a strong partner to the world



MEETING THE MOMENT

LEVERAGING EARTH OBSERVATION (EO) FROM DIPLOMACY TO UKRAINE

"We're facing new times – times that demand we process more information and do so faster. Earth Observation is a strategic tool to solve real challenges in security, defence, and sustainability. These times require us to align, act and accelerate new policies together with research and innovation initiatives."
— Manuel Heitor

STRENGTHENING EUROPE'S DIPLOMATIC ACTION THROUGH DEVELOPMENT COOPERATION

Europe can lead strategic development cooperation, using EO-powered evidence to maximise efficiency, trust, and long-term benefits for partner nations.

TACKLING WATER SCARCITY AND RESERVOIR DEPLETION

EO solutions can be a transformative tool for optimising water use (incl. for irrigation purposes), identification of leakages, and strategic planning for water scarcity response.

BOOSTING PRODUCTION AND PROFITABILITY OF A WATER-SMART AGRIFOOD SECTOR

Precision EO solutions can optimise the whole food value chain. Smart EO-driven policies can secure a resilient, sustainable, and profitable future for the agricultural sector.

RECONSTRUCTING UKRAINE TOWARDS A STATE-OF-THE-ART & SUSTAINABLE INFRASTRUCTURE

EO-driven insights enable a green and secure rebuilding of conflict zones, transforming post-war recovery into a model for future-ready European infrastructure.

ENABLING CHEAPER, EFFECTIVE & RESILIENT URBAN PLANNING

EO integration in urban planning policies strengthens resilience, sustainability, and connectivity. EO can help shape safer and smarter European cities for today and tomorrow.

SECURING ENERGY, COMMUNICATIONS AND TRADE INFRASTRUCTURES IN THE ARCTIC & BALTIC

As Arctic access expands and tensions in the Baltic increase, EO can support Europe in securing vital energy and communication infrastructure and trade routes. This approach would strengthen Europe vis à vis geopolitical trends.

REVERSING EUROPE'S BIODIVERSITY LOSS AND ECOSYSTEM DEGRADATION

EO-enabled applications and data platforms can inform decisions and actions to restore ecosystems, safeguard biodiversity, and secure Europe's natural capital for future generations.



INTEGRATING EO WITH AI FOR ACCELERATED DIGITAL SOLUTIONS

EO provides diverse, real-time data that, when integrated with AI, expands its innovative applications far beyond traditional uses. AI processes EO datasets into actionable insights, enabling Europe to pioneer scalable solutions and strategic foresight across all policy domains with unprecedented speed and accuracy.

WITH LASTING IMPACT

AMPLIFY VALUE ACROSS THE POLICY SPECTRUM

ENERGY

Safeguarding energy transportation routes.
Securing oil & gas supply chains.
Deploying renewables and electrifying transport networks.
Monitoring consumer habits & building efficiency.
Distributing, expanding & modernising the energy system.

FOOD & WATER

Ensuring a transparent & safe food value chain.
Applying precision agriculture & managing livestock.
Developing rural areas & attracting investment.
Planning land use & improving soil quality.

"The question I ask myself is:
How can we valorise this huge untapped potential of Earth Observation?"
— Grammenos Mastrojeni

DIGITAL TRANSITION & GOVERNMENT SERVICES

Building smart & resilient cities.
Ensuring government transparency & reducing bureaucratic burden.
Increasing digital coverage of remote areas.
Developing digital twins & training AI tools.

PEACE & SECURITY

Protecting critical infrastructure & establishing early warning systems.
Detecting illegal trafficking & strengthening border security.
Enhancing military operations & training.
Combatting disinformation threats.
Monitoring treaty negotiation & compliance.

CLIMATE RESILIENCE & EMERGENCY MANAGEMENT

Addressing water scarcity & quality degradation.
Mitigating urban heat islands.
Reducing spread of diseases, ecosystem degradation & habitat stress.
Managing wildfires, floods & severe weather events.
Combatting soil degradation & desertification.

DIPLOMACY

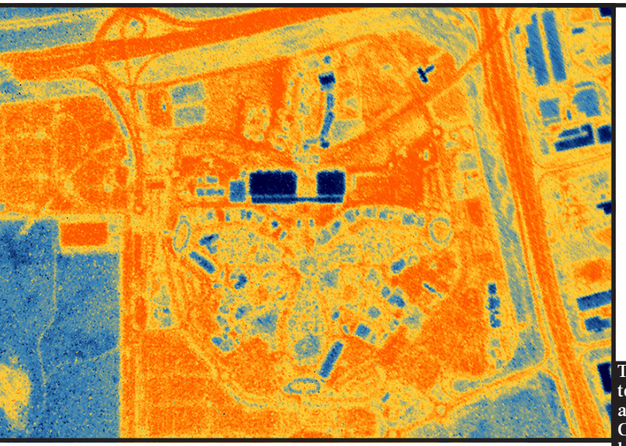
Preventing conflict.
Attracting foreign investments.
Upholding human rights globally.
Ensuring evidence-based development cooperation.

ENERGY

Earth Observation empowers the energy sector, offering verifiable and reliable data to support the transformation and modernisation of Europe's energy systems. Whether planning or monitoring, Earth Observation enables smarter, more secure decisions at every stage of the energy transition.

Space-derived insights can also simplify and optimise reporting practices, reducing administrative burden while enhancing accountability and oversight.

EFFICIENCY · AFFORDABILITY · RESILIENCE



Ensuring energy efficiency by identifying heat loss and detecting areas impacted by the urban heat island effect. By measuring how effectively surfaces in built environments and rural areas absorb and retain heat, thermal satellite imagery supports targeted mitigation and adaptation strategies.

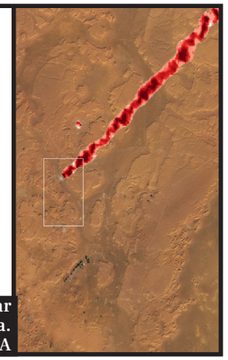
Thermal image of Expo City Dubai, UAE, shows cooler temperatures in the Exhibition Center (two deep blue structures) and within the venue (smaller blue patches), which hosted COP28 formal negotiations, while roads and paved areas are hotter than the surrounding desert (dark orange). HotSat data (2023), processed by SatVu

"I want what a good society wants: dependable energy that keeps Europe clean and innovative. Earth Observation can help make the energy transition possible — and make our energy systems more resilient and effective." — Massimo Tavoni

"Data is important to make our energy system more secure, sustainable, and affordable. Earth Observation has already proven to be useful in many other sectors, so we should also start using its full potential in the energy sector." — Leonardo Meeus

Ensuring the safety and resilience of critical energy infrastructure by monitoring gas and oil pipelines, and detecting leaks through integrated Earth Observation and satellite communications solutions.

Combined Copernicus Sentinel data enable daily detection of significant methane emissions, supporting rapid identification of potential leak sites and enhancing real-time decision-making for pipeline integrity and environmental compliance.



Detection of a methane leak near the Hassi Messaoud oil and gas field, Algeria. Copernicus Sentinel data (January 2020), processed by ESA



Advancing REPowerEU objectives and supporting Europe's energy independence by mapping existing solar panel coverage, identifying untapped solar potential, calculating available roof area for rooftop panels, and feeding data into AI systems to monitor compliance with the EU Solar Rooftop Initiative.

Rooftop solar panels deployed in Amsterdam, the Netherlands. WorldView data (2022), processed by Maxar Technologies

Accelerating the development of renewables by monitoring geothermal physical systems, tracking geothermal power plant development, and detecting geothermal anomalies.



Olkaria IV geothermal power plant's hot water pumping areas in Hell's Gate, Kenya. Pléiades Neo data (2022), processed by Airbus

FURTHER AVENUES AND ADDITIONAL OPPORTUNITIES FOR INCREASING THE BENEFITS OF EARTH OBSERVATION

Enhancing safety of energy transport routes
Ensuring the resilience of energy supply chains
Accelerating the deployment of renewable energies
Integrating distributed energy resources
...

FOOD & WATER

Earth Observation enables reliable, assured and equitable management of natural resources, strengthening food systems and water supply across countries, regions and sectors. When combined with the ongoing AI revolution, satellite insights unlock farmer-friendly services driving tangible improvements in productivity and resilience to climate stressors.

AVAILABILITY · QUALITY · SAFETY

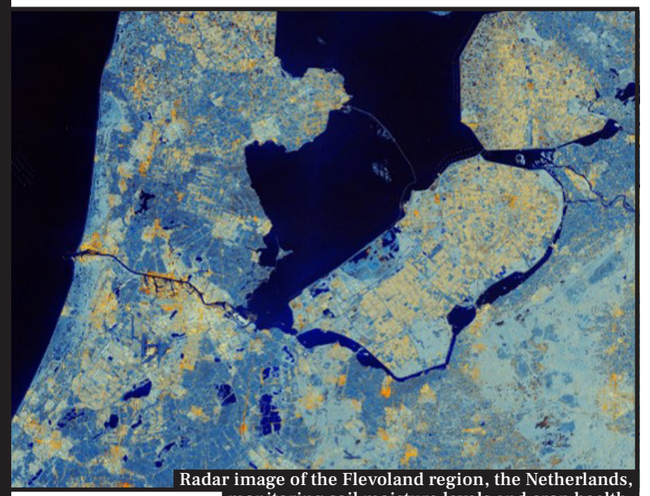


Soil Organic Carbon estimations based on spectral analysis of Bare Soil Composites in the Czech Republic. Copernicus Sentinel and Landsat data (2020-2023), processed by ESA

Pioneering Earth Observation applications to support the European Union's Common Agricultural Policy monitoring efforts, especially the new eco-schemes designed to reward climate- and nature-friendly farming. Developed through collaboration between ESA, European EO companies, and National Paying Agencies, advanced monitoring of soil organic carbon for carbon footprint calculations ensures accurate and efficient allocation of subsidies and incentives.

"Geospatial technologies are among the most promising technologies for the agrifood sector." — Vincent Martin

"To feed the growing global population and eradicate poverty, we need to double global food production by 2050 on the same land area as today, using half as much water and energy. This requires technology and incentives for farmers to produce more with less." — George Freeman



Radar image of the Flevoland region, the Netherlands, monitoring soil moisture levels and crop health. Copernicus Sentinel data (2024), processed by ESA

Helping local authorities and regional actors to shape efficient water management systems through information on soil moisture, water availability and crop health, complementing and optimising in-situ inspections.



Crop classification identifying crop types in Flevoland, the Netherlands, such as cereal (purple), potato (brown), sugar beet (blue), maize (orange), onion (ochre), tulips (red), other (pink). Green indicates grasslands. Copernicus Sentinel data (2022), processed by ESA

Earth Observation and Artificial Intelligence combined allow precise and automated detection of crop types, informing farmers' needs for fertilisers, while improving productivity, supporting yield estimation, and reducing costs.

FURTHER AVENUES AND ADDITIONAL OPPORTUNITIES FOR INCREASING THE BENEFITS OF EARTH OBSERVATION

Informing rural development
Supporting implementation of farming laws
Increasing biodiversity
Ensuring traceability and safety of the food value chain
...

DIGITAL TRANSITION & GOVERNMENT SERVICES

Earth Observation enhances the efficiency, transparency and credibility of public administration, supporting better decisions, cost savings, and improved service delivery, from spatial planning to tax collection, with verifiable data combatting misinformation.

Available end-to-end solutions can further strengthen Europe's digital sovereignty by ensuring institutions can increasingly rely on native and trusted digital backbones.

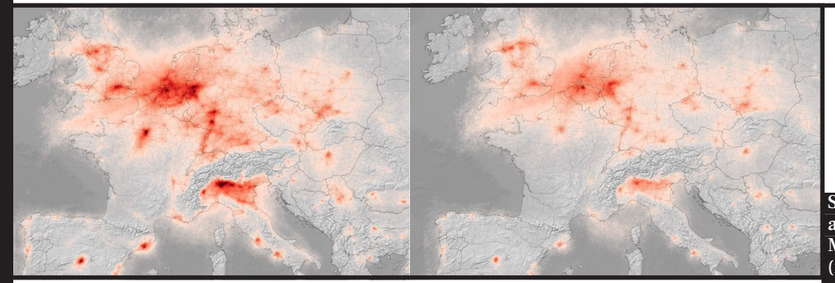
TRUST · EFFICIENCY · DELIVERY



Building footprint and height, from low (2 meters, red) to high (7.4 meters, green), Kigali, Rwanda. Pléiades Neo data (2017), processed by The World Bank

Transforming tax collection systems by using Earth Observation data to automatically estimate total floor space by integrating building height with footprint data. Combined with spatial information and land price data, EO data inform the processes of computer-assisted mass appraisal (CAMA) used by government agencies to help assess property valuation for tax collection.

"Our challenge is to build a truly sovereign EuroStack — one that links Earth Observation with democratic data governance, industrial competitiveness, and the defence of the European model. Space must serve not only innovation, but also strategic autonomy and the public good." — Francesca Bria

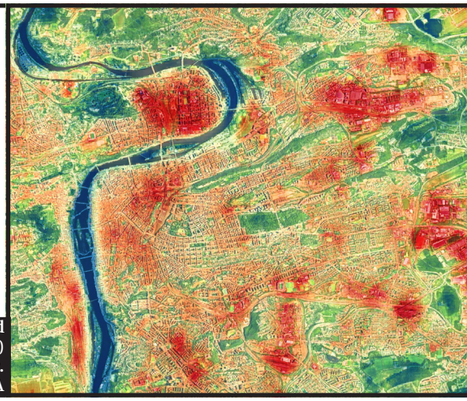


Visibly confirming reduced emissions from mobility and demonstrating the impact of policy decisions in times of crises, such as lockdowns during pandemics.

Significant decrease in nitrogen dioxide levels across Europe during COVID-19 lockdowns, March-April 2019 (left) and March-April 2020 (right). Copernicus Sentinel data (2020), processed by ESA

"Through Earth Observation, we can deliver real societal benefits and welfare in a time of competing interests and budgetary needs." — Ioana Petrescu

Maps that jointly leverage Earth Observation data and Artificial Intelligence highlight the most heat-vulnerable locations in cities, thus justifying public adaptation measures for cooling, such as green roofs, misting systems, tree canopy cover, and drinking fountains. Collaborative efforts between ESA and NASA help to advance land surface temperature monitoring, improving the precision of urban heat assessments.



High-resolution Earth Observation-derived heat vulnerability map of Prague, showing hot surfaces (red) and cooling effects (green) of parks, vegetation, and water. NASA ECOSTRESS data (2022), processed by ESA

FURTHER AVENUES AND ADDITIONAL OPPORTUNITIES FOR INCREASING THE BENEFITS OF EARTH OBSERVATION

Enhancing industrial efficiency
Building energy-efficient digital infrastructure
Expanding digital coverage to remote areas
...

PEACE & SECURITY

Earth Observation is a key enabler of sovereignty in the 21st century, providing insights into fragile contexts, detecting early signs of instability, protecting society and assessing the impact of conflicts on people and infrastructure.
It enables more precise attribution, stronger demand for accountability, and more assertive response, strengthening peace and security through facts, rather than assumptions and disinformation.

DETERRENCE · PROTECTION · STABILITY



Demolishing of Kakhovka dam causing flooding (red),Ukraine.
Copernicus Sentinel data (May-June 2023), processed by Geospatial Institute, Austrian Ministry of Defence

Guiding rapid disaster response, recovery, and safe manoeuvring through mapped devastated areas after infrastructure collapse.

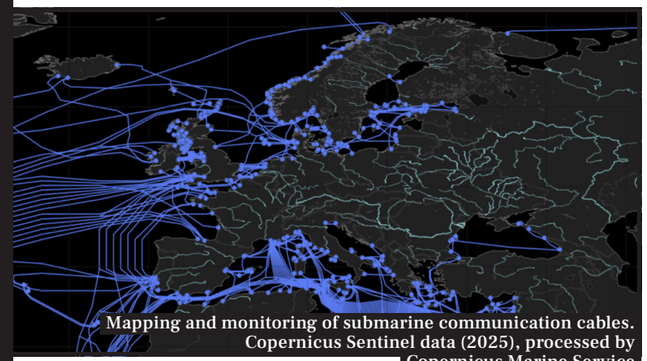
"When considering a truce or ceasefire, relying on an independent understanding of the situation on the ground in real time is essential—especially when evaluating the feasibility of troop deployment." — Claude-France Arnould



Mass grave site identified in Bucha, Ukraine, following the withdrawal of Russian troops.
Worldview data (2022), processed by Maxar Technologies

"We need to anticipate threats, detect hostile behaviour, and attribute it. The clearer the attribution, the better the response from countries concerned." — Fabrice Pothier

Documenting war crimes and ensuring accountability by identifying damage to civilian infrastructure and providing objective evidence of targeted attacks in conflict zones.



Mapping and monitoring of submarine communication cables.
Copernicus Sentinel data (2025), processed by Copernicus Marine Service

Continuous and near-real-time monitoring of marine environments to enable early detection of anomalies or disturbances to submarine communication cables. Public-private partnerships facilitate access to commercial radio frequency detection services, helping to safeguard these critical infrastructures.



Illegal oil shipment from Russia to North Korea, a violation of UN sanctions.
Planet Labs data (November 2024), processed by Open Source Centre

High-resolution satellite imagery combined with Automatic Identification System (AIS) data for precise detection and attribution of vessels, enabling states to detect treaty violations at sea.

FURTHER AVENUES AND ADDITIONAL OPPORTUNITIES FOR INCREASING THE BENEFITS OF EARTH OBSERVATION

Strengthening pandemic response and health crisis management
Preventing conflicts
Ensuring border security
Combatting illegal drug and human trafficking
…

CLIMATE RESILIENCE & EMERGENCY MANAGEMENT

Earth Observation provides timely information to support rapid decision-making in emergency situations, moving beyond monitoring to enable real-time intervention when it matters most. In an era of interacting and compounding crises, EO can power systems that help us to see the bigger picture, act with precision, and build long-term resilience.

PREVENTION · RESPONSE · RECOVERY



Digital models of coastal cities and landscapes for testing urban planning and resilience strategies, ensuring disaster preparedness and evaluating success of interventions.

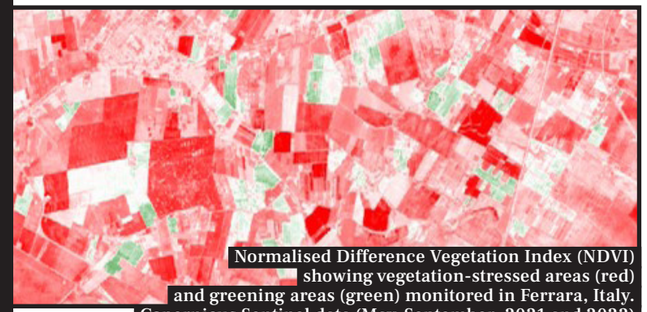
Flooded streets and low-lying areas in Tonga's Digital Twin, showing simulated water depth and extent during scenarios of 0m sea level rise (top) and 4m sea level rise (bottom). Pléiades Neo data (2024), processed by Space Data

"We need to use everything at our disposal – the right tool in the right situation and build new ones to meet the increasing demands of emergency management." — Fabrizio Curcio



Forest loss (in red) between 2018 and 2023 in Suceava County, Romania. Copernicus Sentinel and Hansen Global Forest Change data (2023)

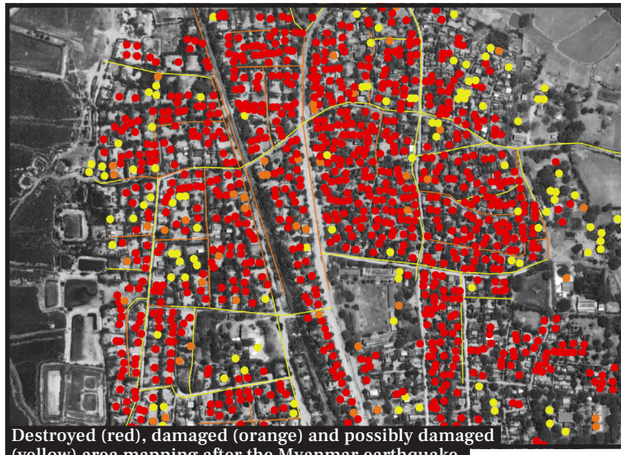
Monitoring forest degradation by detecting tree cover loss and surface change in protected woodland areas affected by resource mismanagement and over-logging.



Normalised Difference Vegetation Index (NDVI) showing vegetation-stressed areas (red) and greening areas (green) monitored in Ferrara, Italy. Copernicus Sentinel data (May-September, 2021 and 2022)

Detecting risks of water scarcity by monitoring vegetation decline and surface changes during droughts in groundwater-dependent agricultural regions.

"In the case of the ocean, many challenges and opportunities converge, from sustainability to security. The oceans are fundamental challenges for multilateral efforts leveraging Earth Observation."
— Geneviève Pons



Destroyed (red), damaged (orange) and possibly damaged (yellow) area mapping after the Myanmar earthquake. Copernicus Sentinel data (March 2025), processed by Copernicus Emergency Management Service

Supporting disaster response by delivering rapid, sustained insights into the needs and risks of impacted areas – even when direct access is limited. Informing humanitarian policy by guiding relief operations, damage assessments, and reconstruction efforts. Collaboration between ESA and the International Red Cross and Red Crescent Movement strengthens this approach by advancing the application of satellite Earth Observation for disaster management and humanitarian aid worldwide.

FURTHER AVENUES AND ADDITIONAL OPPORTUNITIES FOR INCREASING THE BENEFITS OF EARTH OBSERVATION

Preventing the spread of diseases and pest outbreaks
Monitoring coastal erosion and sea level rise
Supporting biodiversity conservation
...

DIPLOMACY

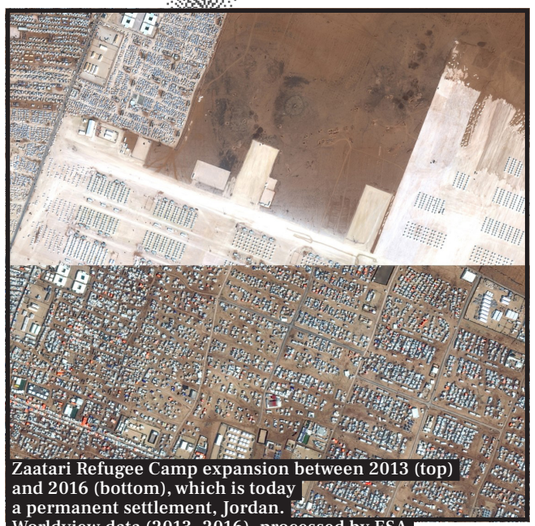
Earth Observation benefits evidence-based negotiations, establishing trust and equipping Europe with clear and reliable insights when tackling the most pressing regional and global challenges of our time, with transversal impact across the domains of energy, food & water, digital transition, peace & security, climate resilience.
Space must be better leveraged as a soft power tool to protect shared values, safeguard strategic assets and resources, offer concrete solutions to global partners, and empower local communities.

INTEGRITY · LEADERSHIP · TRUST



Monitoring of checkpoint activity between Ethiopia and Sudan, Doka, Sudan.
CNES/Airbus data (2018)

Detecting early signs of military activity in conflict-affected areas and essential borders through satellite imagery, enabling fact-based diplomacy, reducing disinformation, and preventing escalation of conflict.



Zaatari Refugee Camp expansion between 2013 (top) and 2016 (bottom), which is today a permanent settlement, Jordan.
Worldview data (2013, 2016), processed by ESA

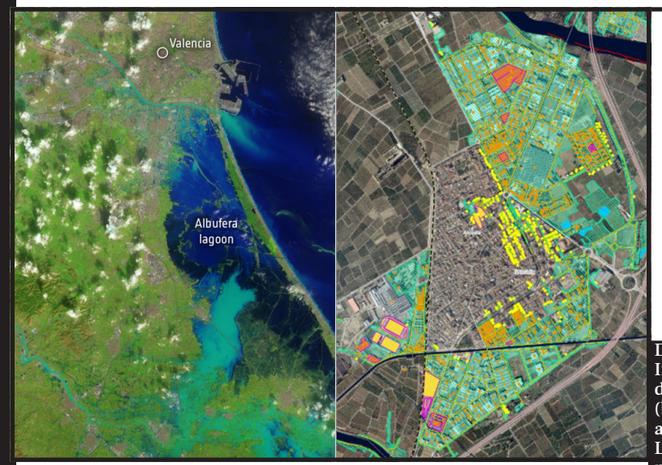
Earth Observation data and value-added services provide strategic support to populations affected by conflicts. Enabling efficient refugee camp planning and sustainable humanitarian aid by identifying safe terrain, guiding flood-resilient drainage design, and monitoring camp expansion for effective site management.

"International cooperation is a soft power tool that Europe can use and rediscover, including through Earth Observation, which is key to fostering consensus and a cooperative approach." — Luigi Narbone

EARTH OBSERVATION TURNS SHARED CHALLENGES INTO SHARED SOLUTIONS, BUILDING THE FOUNDATION FOR TRUST AND COOPERATION

Earth Observation enables countries to identify shared challenges, from water scarcity and ecosystem degradation to unregulated land use and shifting population pressures.

By providing neutral, cross-border data, EO lays the foundation for international cooperation and treaty design. It helps to define the "where" and "why" of mutually beneficial action.



Devastating floods in Valencia, Spain (left), triggered the International Charter "Space and Major Disasters", enabling damage analysis in Alcudia (right). The map shows flood trace (blue), damaged buildings (orange dots), facilities (orange stripes), and possibly damaged roads (yellow).
Landsat data (2024), processed by ESA; Pléiades data (2024), processed by Copernicus Emergency Rapid Mapping Service

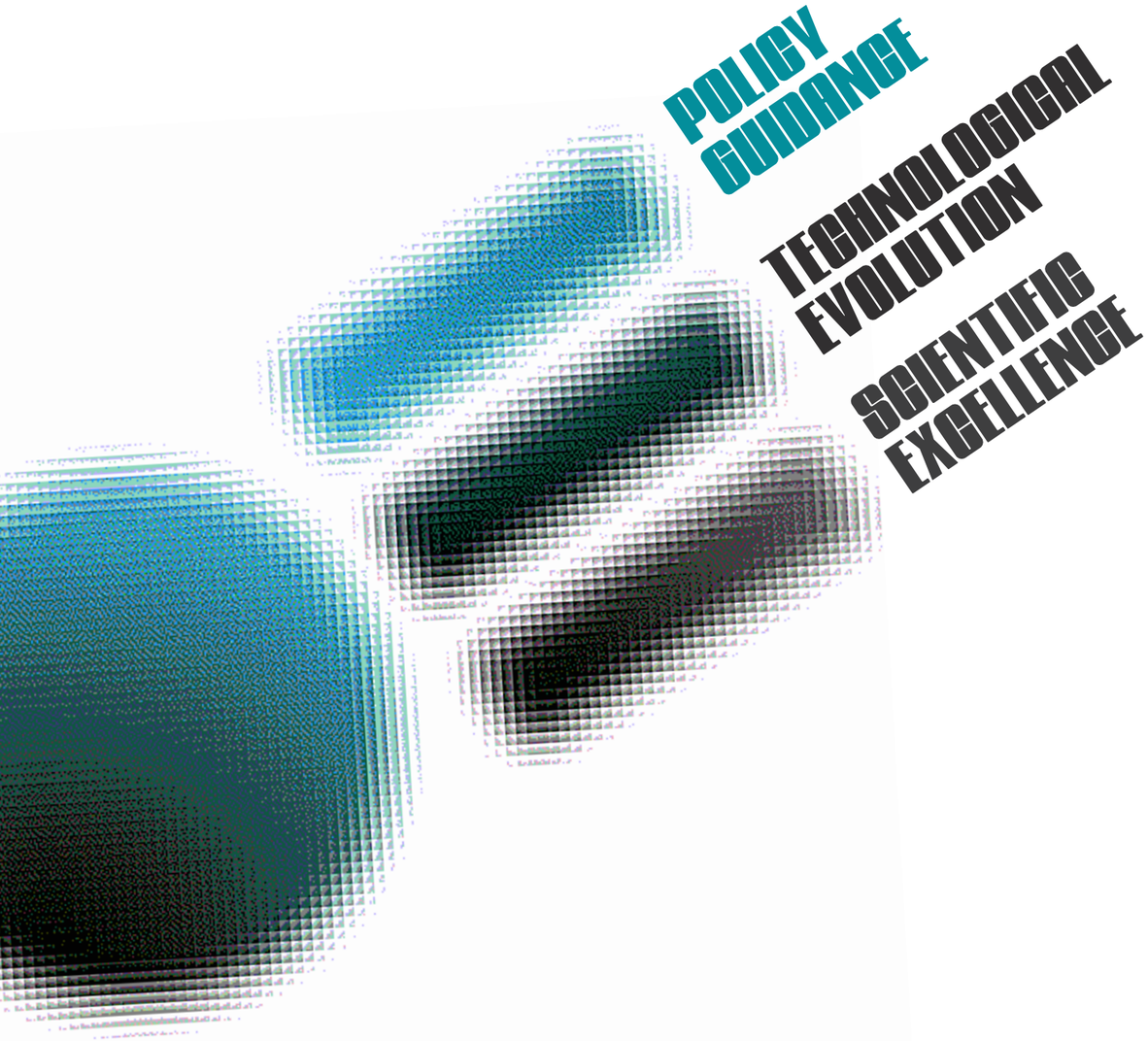
Strengthening international cooperation through interoperable information services and Earth Observation data shared among space agencies, commercial satellite companies, emergency services, and humanitarian actors, enabling rapid flood extent mapping and water detection. The International Charter "Space and Major Disasters", initiated by ESA and CNES, plays a key role in mobilising satellite resources to support global disaster response efforts.

"Satellites, paired with supercomputing and artificial intelligence, must become instruments of global water diplomacy – helping us predict, manage, and safeguard water resources where they are needed most. This is a call to build a bold global partnership, because access to clean, safe, and sustainable water is a shared responsibility – and a foundation for global stability and peace." — Robert Golob, Prime Minister of Slovenia

FURTHER AVENUES AND ADDITIONAL OPPORTUNITIES FOR INCREASING THE BENEFITS OF EARTH OBSERVATION

Facilitating development cooperation
Supporting rescue operations
Countering disinformation threats
Informing treaty negotiation and monitoring compliance
...

THE VISION



UNLOCKING IMPACT

For Earth Observation to deliver on its full potential, scientific excellence and technological innovation need to be complemented by clear policy guidance, turning capability into results.

SMARTER POLICYMAKING

Earth Observation offers powerful tools to enable better policy design, implementation and evaluation. Governments can deliver assertive, cost-efficient and coherent impact by increasingly relying on the outcomes of space programmes.

A LIVING PLATFORM

Earth Observation for policy is much more than just delivering information. Maintaining regular channels for policy input is essential to guiding and shaping space activities. It ensures technology development stays agile, relevant, and aligned with rapidly evolving priorities, keeping Europe on the front foot.

FOOD, WATER, ENERGY, DEMOCRACY, PEACE
EMPOWERED THROUGH SPACE

MEANINGFUL INSIGHTS

AGENDA SETTING & POLICY FORMULATION

Detecting emerging trends in their early stages and guiding in-situ data collection

Simulating future scenarios with accuracy thanks to user-friendly digital twins based on Earth Observation data

Combining socio-economic data with insights from satellites to build the interdisciplinary basis needed for policy and lawmaking

"To set a proactive policy cycle, innovation through the use of Earth Observation data is essential." — Chris Skidmore

POLICY MONITORING & ASSESSMENT

Precisely measuring the outcomes of policy decisions and showing success through time-series analysis and integration of Artificial Intelligence

Measuring, reporting and verifying compliance, uncovering unintended consequences and identifying external factors

Gathering relevant insights for updating policies

"You can't manage what you don't measure. Earth Observation needs to be fully absorbed into policy cycles." — Niklas Nienaß

EVERY STEP OF THE WAY

POLICY ADOPTION & IMPLEMENTATION

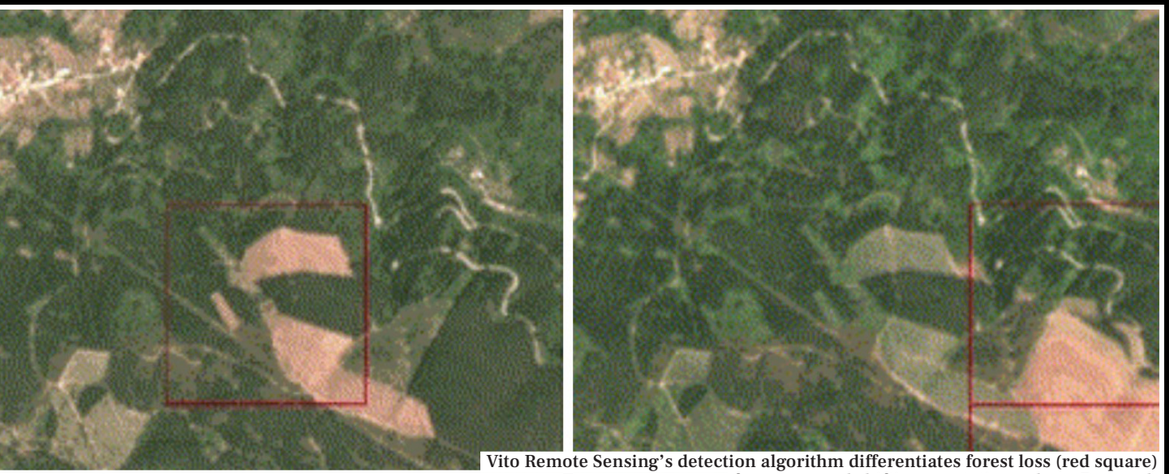
Supporting evidence-based decision-making with detailed and comparable data over time and locations

Swiftly developing implementation plans building on Earth Observation-powered infographics

Optimising and dynamically adjusting resource allocation, and spotting inefficiencies through geospatial intelligence and predictive analysis

AGENDA SETTING & POLICY FORMULATION

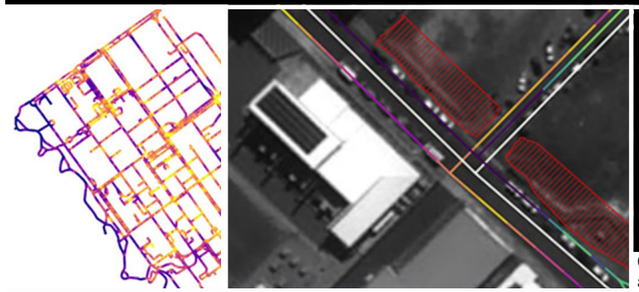
ASSESS RISKS · SET GOALS · IDENTIFY SOLUTIONS



Vito Remote Sensing's detection algorithm differentiates forest loss (red square) from seasonal shifting vegetation colour, Portugal. Copernicus Sentinel and Planet Fusion data (June-July 2019, top; June-July 2020, bottom), processed by Vito Remote Sensing

By detecting land cover changes over time and leveraging deep neural networks, Earth Observation-based insights can distinguish genuine changes like deforestation from seasonal variations and measure the real scale of the issue at stake, strategically setting the agenda for **land-use policies**.
Trained on extensive satellite imagery, these tools provide accurate and timely detection, enabling policymakers to prioritise risks, justify early action, and drive effective policy formulation.

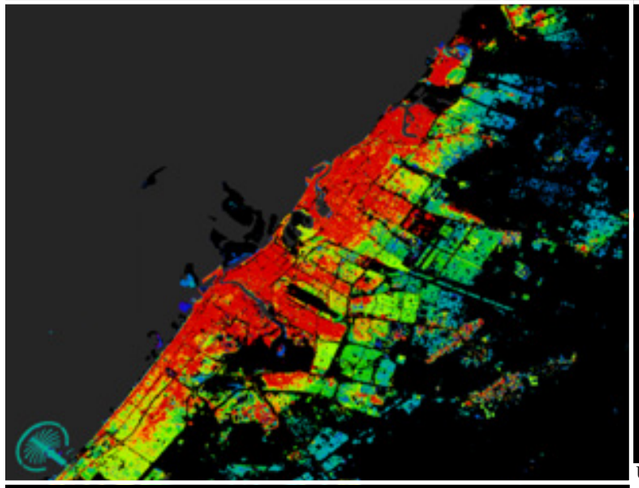
"Earth Observation isn't just an add-on or a technical tool. It must be embedded at the core of policy design."
— George Papaconstantinou



Guiding **urban greening policy** formulation through satellite imagery and AI-powered insights to identify high-priority areas for tree planting by detecting where heat and tree stress intersect with low canopy cover.

Context-aware heat stress index (left) and two high-priority areas for tree planting (right) in Darwin, Australia, identified through SatVu processed satellite imagery and the Office of Planetary Observations (OPO) AI-driven Nature Data platform. HotSat data (2023), processed by SatVu

COPERNICUS DATA ENHANCES RISK ASSESSMENT AND RESOURCE ALLOCATION, WITH THE EUROPEAN MARITIME SAFETY AGENCY (EMSA) INDICATING OVER 80% COST SAVINGS IN OIL SPILL MONITORING AND RESPONSE TIMES REDUCED BY MORE THAN 24 HOURS IN MARITIME EMERGENCIES.



When urban expansion outpaces traditional mapping, high-resolution Earth Observation data provide local governments with timely insights into infrastructure changes, identifying high-risk areas and formulating effective **housing policies**.

In regions with rapid urban expansion and environmental risks, collaborations between ESA, DLR and Google Earth Engine have enabled the precise monitoring of human settlements, informing sustainable urban development and strategic planning

Urban expansion of Dubai, United Arab Emirates, mapped as part of the World Settlement Footprint Evolution throughout the 1980s (red-orange), the 1990s (yellow-light green), the 2000s (dark green-light blue) and the 2010s (dark blue-purple). Landsat data (1985-2015), processed by DLR

POLICY ADOPTION & IMPLEMENTATION

TAKE DECISIONS · TAILOR RESPONSE · OPTIMISE RESOURCES



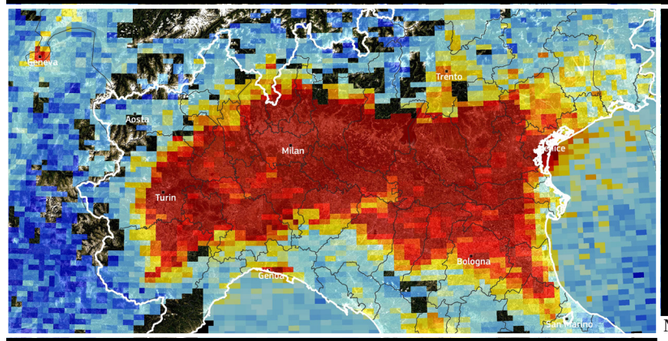
Enabling more accurate and transparent implementation of **carbon credit policies** by mapping land cover across forest carbon project areas before and during implementation, thus verifying environmental impacts through Earth Observation data.

Integrating Earth Observation data into carbon credit systems ensures accurate baseline assumptions, estimates carbon sequestration potential, reduces manual auditing costs, builds stakeholder trust, and attracts investment for meaningful climate action.

Proposed area of a mangrove forest (solid white line) and its leakage belt (dotted white line) in Sembilang, Indonesia, where mangroves appear in dark green and plantations in red. Copernicus Sentinel, Landsat and ALOS PALSAR data (2023), processed by Space Intelligence

THE SPANISH CIVIL GUARD (GUARDIA CIVIL) REPORTS OVER 24 HOURS SAVED PER INVESTIGATION WHEN USING COPERNICUS DATA ACROSS SEVERAL OPERATIONAL UNITS (E.G. ENVIRONMENTAL PROTECTION, RISK ANALYSIS), RESULTING IN MORE EFFICIENT RESOURCE USE.

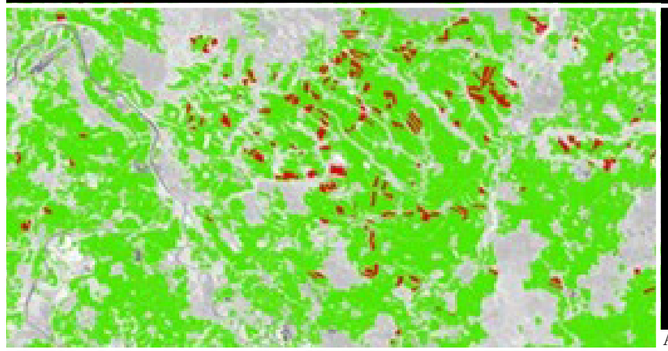
> ***"Robust data from space are key in order to move cities into the 21st century."*** — Diana Ürge-Vorsatz



Supporting the adoption of **policies to reduce air pollution in cities** through high-resolution granular data on air quality, translating policies to the regional and local level and enabling targeted interventions.

Nitrogen dioxide levels detected over northern Italy, with a visible concentration around Milan. Copernicus Sentinel data (2021)

THE ITALIAN AGENCY FOR PAYMENTS IN AGRICULTURE (AGEA) HIGHLIGHTS 30-40% COST SAVINGS FROM USING EARTH OBSERVATION DATA FOR FARM INSPECTION AND MONITORING, ALONGSIDE TIME SAVINGS OF OVER 2 DAYS PER AVERAGE FIELD INSPECTION.

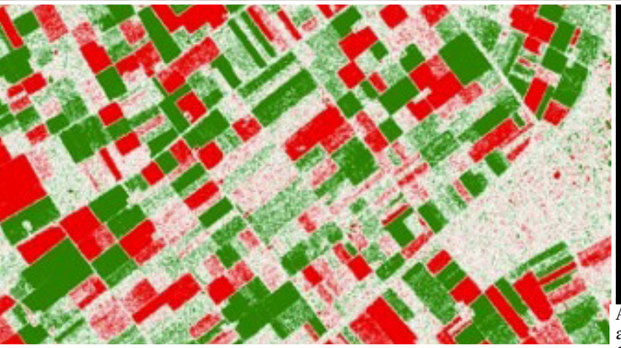


Accelerating **EU's Common Agricultural Policy (CAP)** implementation by monitoring compliance and saving resources through continuous, large-scale observation of crop patterns across more than a million parcels in near real time. Combined optical and radar data from Copernicus Sentinel satellites help to monitor compliance with anti-erosion rules by detecting prohibited crops on erosion-prone land.

Analysis of parcel compliance with anti-erosion rules - compliant (green) and non-compliant parcels (red). Copernicus Sentinel data (2023), processed by ESA

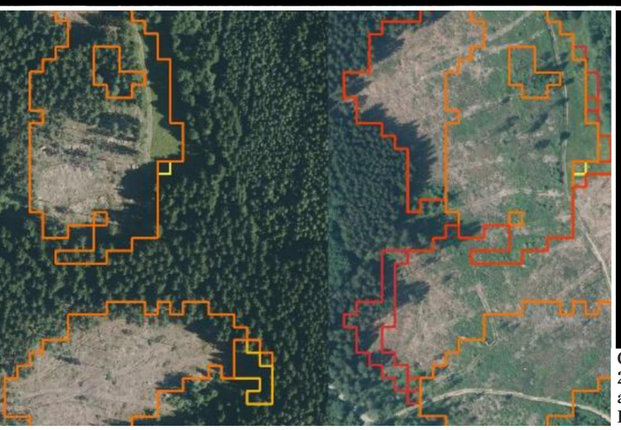
POLICY MONITORING & ASSESSMENT

SET INDICATORS · VERIFY COMPLIANCE · REVIEW POLICIES



Earth Observation data keeps track of vital indicators like soil moisture and crop health in rural landscapes, revealing where **agricultural policies** have succeeded or where land and vegetation degradation signals the urgent need for reform.

Agricultural land monitoring showing improving (green) and declining (red) farmland conditions, Flevopolder, the Netherlands. Copernicus Sentinel data (2023-2024)



Wide-scale comparative analysis and long-term monitoring of forest density provide historical baseline data, enabling the training, testing, and validation of remote sensing systems for forest damage detection at a national level.

This facilitates automated, standardised nationwide identification of forest damage and utilisation areas, crucial for evidence-based evaluation of **forest policy** effectiveness.

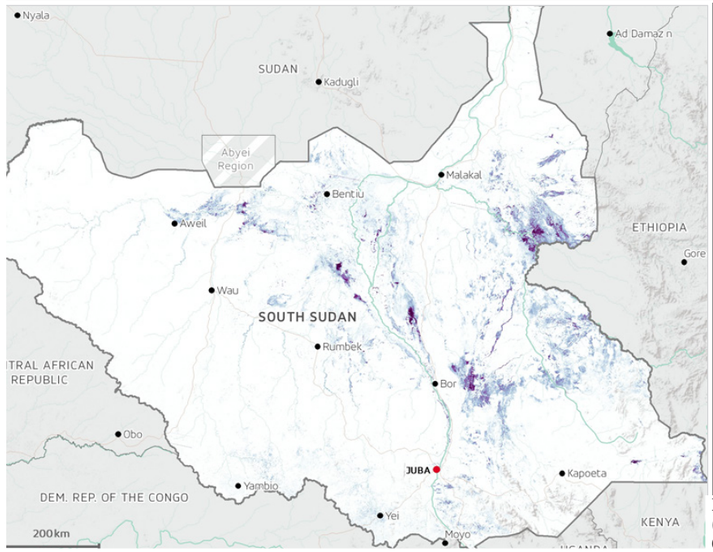
Growing forest perturbation patches detected in 2018 (yellow), 2019 (light orange), 2020 (dark orange), 2021 (red), and 2022 (dark red) across Germany. Pléiades, WorldView, PlanetScope, and aerial data (2022), processed by GeoSN

> ***"Only few EU Member States use precision farming enabled by Earth Observation in their CAP eco-schemes to subsidise farmers - we need to increase our support to precision agriculture."*** — Janusz Wojciechowski



Remote sensing allows for the comparison of reforested areas with undisturbed control zones. Integrated with AI models, Earth Observation enhances the monitoring of **reforestation policy** outcomes by assessing post-wildfire restoration efforts and understanding how climate variability influences forest recovery.

Vegetation recovery surveillance in Sierra Bermeja, Spain, following a devastating wildfire in 2021. Copernicus Sentinel data (2024), processed by Copernicus Emergency Management Service



By delivering essential data to assess the vulnerability of infrastructure, identify gaps, and predict outcomes of potential changes, Earth Observation helps evaluate the effectiveness of **anti-flood policies** and determine where adjustments are needed.

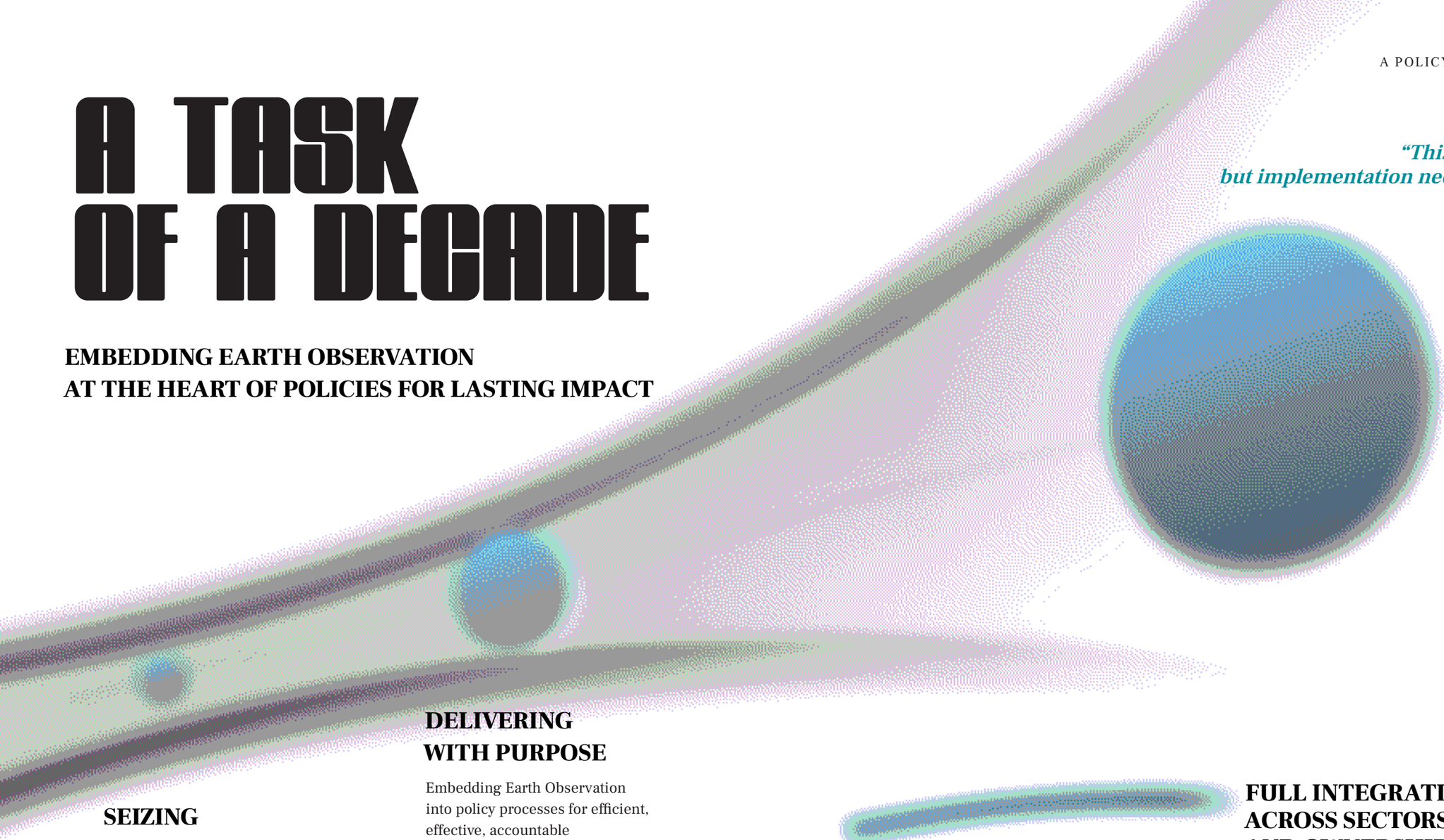
Collaboration between ESA and International Financial Institutions (IFIs) enhances flood management strategies by tracking the extent, frequency, and duration of heavy precipitations to produce detailed maps of flood hazards.

Flood frequency mapping in South Sudan showing high (dark purple) and low (light blue) flood frequency areas. Copernicus Sentinel data (2015-2022), processed by ESA

A TASK OF A DECADE

EMBEDDING EARTH OBSERVATION AT THE HEART OF POLICIES FOR LASTING IMPACT

"This vision comes at the right time, but implementation needs to start now." — Klaus Welle



ACHIEVING THE STRATEGIC ADVANTAGE

Strengthening autonomy in setting objectives and managing the means to attain them for positioning Europe as a strong, reliable global partner.

DELIVERING WITH PURPOSE

Embedding Earth Observation into policy processes for efficient, effective, accountable decision-making and action.

SEIZING IMMEDIATE OPPORTUNITIES

Leveraging Earth Observation in response to pressing challenges by maximising the use of existing solutions.

NOW

CONTINUOUS POLICY GUIDANCE

SUSTAINED DIALOGUE

FULL INTEGRATION ACROSS SECTORS AND OWNERSHIP OF SOLUTIONS BY POLICY COMMUNITIES

2030+

BRIDGING THE SPACE AND POLICY COMMUNITIES

THE JOURNEY STARTS TODAY

BUILDING ON EUROPE'S RICH HERITAGE IN EARTH OBSERVATION (EO)

Bridge the gap between EO and policy needs through a dedicated platform for accessing solutions.

Accelerate EO data sensing & delivery through new European integrated solutions with global 24/7 availability.

Define a European approach to market creation and shape demand for EO based on strong public services & governance.

Deliver solutions with a focus on local policies, both in cities and rural areas.

Boost capacity-building networks targeting specific policy domains with transversal approaches.

Develop the full spectrum of critical EO capabilities to ensure European security and national sovereignty through existing and new ESA, EU and national programmes.

SPACE

Create structured entry points in ESA and EU space programmes to integrate policy guidance.

Promote storytelling that showcases everyday reliance on EO services.

"Earth Observation data are key strategic tools for implementing European and international policies. Our satellites and solutions must be integrated into long-term policy vision." — Simonetta Cheli

POLICY

Mobilise agriculture, environment, digital, defence, foreign relations budgets to scale demonstrated EO solutions.

Embed direct incentives for uptake of EO through regulatory action and targeted support schemes.

Launch a public-private procurement framework to scale mature EO services across Europe.

Identify focal points across policy institutions for liaising with the space community.

Consolidate smaller and dispersed funding schemes for R&D into larger, impact-driven envelopes.

Leverage the speed and value of commercial solutions and procurement schemes to unlock faster cycles from idea-to-market.

Ensure a continuous innovation pipeline with accelerated and scaled space programme development for European leadership in EO.

Leverage the ESA Council at Ministerial Level and the MFF proposal, to substantiate Europe's declared policy ambitions.

Ease information access with user-friendly and intuitive applications tailored to diverse policy actors.

Reduce the time from creation to adoption by designing solutions with policy communities from the outset.

Boost commercial EO solutions through policy incentives and targeted regulatory alignment.

There is a wider meaning behind Earth Observation: How did we miss what was happening out there?
— Rolf Alter

THE VISION DEVELOPMENT PROCESS

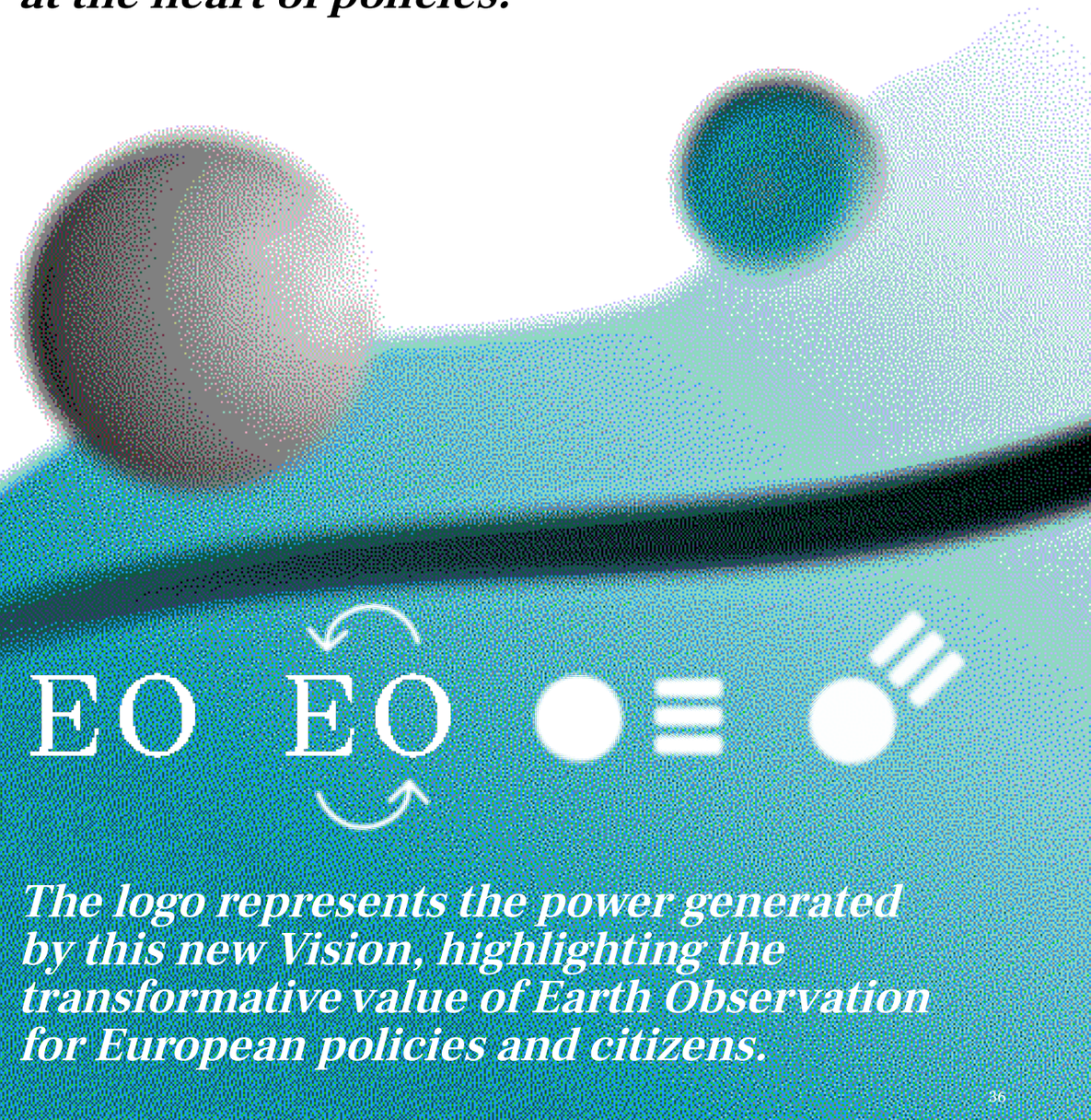
This Policy Vision for Earth Observation was developed by ESPI between September 2024 and April 2025, in collaboration with the European Space Agency (ESA). The Vision is grounded in an analysis of the priorities derived from over 300 policy documents and 1200+ Earth Observation projects. This document highlights selected policy domains and related objectives that emerged from the analysis. The research was complemented by a thorough consultation campaign with over 50 European stakeholders from different policy domains.

This Vision could not have come to life without a collective effort reaching far beyond ESPI's walls. ESPI and ESA would like to extend their gratitude to the participants of the high-level policy workshop, who gathered at the Royal Society in London on 3 April 2025 to provide their precise inputs and strategic guidance.

The Vision will be brought by ESPI to all relevant policy communities as an input to foster a closer dialogue with the space sector. ESPI is committed to facilitating this dialogue together with a network of policy and space stakeholders towards the implementation of concrete solutions to the issues at stake.

It's time to open up to new voices! Where policy needs are most acute, space must follow even into unmapped terrain.

We call on European policy and thought leaders to guide, shape, and integrate Earth Observation at the heart of policies.



The logo represents the power generated by this new Vision, highlighting the transformative value of Earth Observation for European policies and citizens.

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Jeremy Wilks

MODERATOR

Science Correspondent at Euronews

WHAT IS YOUR ROLE IN BRINGING THIS VISION TO LIFE?

BE A CATALYST IN YOUR COMMUNITY

2025
concept & development by the European Space Policy Institute (ESPI)
design by (a+b) annalisa dominoni, benedetto quaquaro